

B: The equivalent car and its mass

In this appendix we shall justify the assertions (4.5) about the properties of *the equivalent car*.

For a start, let us recall how to define the moment of inertia I for the merry-go-round around its axis. Thinking of it as a collection of masses Δm_i , each having a well defined distance r_i from the axis, one has

$$I = \sum_i \Delta m_i r_i^2 \quad (\text{B1})$$

The first thing we are going to prove is this: The mass determined in the two experiments in fig. 5.2 is the mass defined by

$$m = \frac{I}{R^2}. \quad (\text{B2})$$

Proof: With each point s on the circular s -axis goes the angle $\theta = \frac{s}{R}$ and the equation of motion for the model point is

$$I \frac{d^2 \theta}{dt^2} = M, \quad (\text{B3})$$

where M is the torque. In terms of m from (B2), this can be written as

$$m \frac{d^2 s}{dt^2} = \frac{M}{R}, \quad (\text{B4})$$

When the model point is suspended between the two springs as in fig. 5.2a, the torque equals $-Rks$ where k is the force constant. Inserting this and $\theta = \frac{s}{R}$, the equation of motion (B4) takes the form

$$m \frac{d^2 s}{dt^2} = -ks, \quad (\text{B5})$$

solved by a harmonic oscillation with the period T given by

$$T = 2\pi \sqrt{\frac{k}{m}} \Leftrightarrow m = \frac{kT^2}{4\pi^2}. \quad (\text{B6})$$

So, the mass we determined in fig. 5.2a is the one defined in (B2) above.

To see that the pendulum experiment in fig. 5.2b determines the same mass, we first note that the extra mass Δm attached to the model point increases the moment of inertia from mR^2 to $(m + \Delta m)R^2$. With the torque of the gravitational pull given by $-\Delta m \cdot g \cdot R \sin \theta$, the equation of motion (B4) becomes

$$-\Delta m \cdot g \cdot \sin \theta = (m + \Delta m)R^2 \frac{d^2 \theta}{dt^2} \quad (\text{B7})$$

For small amplitudes ($\sin \theta \simeq \theta = \frac{s}{R}$) this can be written as

$$(m + \Delta m) \frac{d^2 s}{dt^2} = -k \cdot s \text{ where } k = \frac{\Delta m \cdot g}{R} \quad (\text{B8})$$

This describes a harmonic oscillation with the period T given by

$$T = 2\pi \sqrt{\frac{k}{m + \Delta m}} \Leftrightarrow m + \Delta m = \frac{kT^2}{4\pi^2}. \quad (\text{B9})$$

Hence

$$m = \frac{kT^2}{4\pi^2} - \Delta m = \left(\frac{g}{4\pi^2} T^2 - 1 \right) \Delta m, \quad (\text{B10})$$

which is exactly the m we determined by the pendulum experiment in fig. 5.2b.

The assertions of (4.5a) and (4.5b) now follows from the following observation: Assume that the model point is acted upon by a force F , directed along the circular s -axis (and signed accordingly). The torque M then equals RF and the equation of motion (B4) takes the form,

$$m \frac{d^2 s}{dt^2} = F. \quad (\text{B11})$$

This is the same as the equation of motion (namely Newton's second law) for a mass m moving on a linear s -axis, acted upon by a force F , directed along this axis.

Yet another result should be verified, namely the postulate in (5.1) that m increases from m_e to

$$m = m_e + m_{\text{model}} + m_{\text{load}}; \quad m_{\text{load}} = \left(\frac{r_{\text{counter}}}{R} \right)^2 m_{\text{counter}} \quad (\text{B12})$$

when the model is mounted in the distance R from the axis, and a counterbalance of mass m_{counter} is mounted in the distance r_{counter} . According to the (B1), these additions mean that I increases by

$$\Delta I = m_{\text{model}} \cdot R^2 + m_{\text{counter}} \cdot r_{\text{counter}}^2 \quad (\text{B13})$$

According to (B2), this means that m increases by $\Delta I/R^2$ which is just the amount aimed at.